1 Summary

Model (Run-name): model1 (test)

Iteration: 41

TAC: 154899.43267562543

Took: 4.328382730484009s

Total Time: 106.02697658538818s

Epsilons:

- ε_{LMTD}: 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area}: 1e-06
- ε_{Beta}: 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

Gamsworld Solution: `http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html`

2 Results

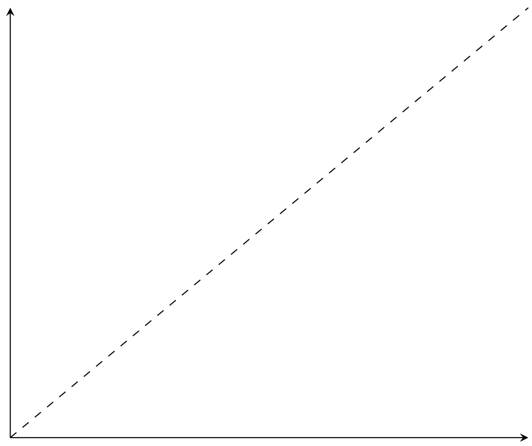
Variable	Value
$z_{1,1,1}$	1.0
$z_{1,2,2}$	1.0
$z_{2,1,2}$	1.0
$z_{cu,1}$	1.0
$z_{cu,2}$	1.0
$z_{hu,1}$	1.0
$A^{\beta}_{1,1,1}$	71.04727671123747
$A^{\beta}_{1,2,2}$	69.0865541528717
$A^{\beta}_{2,1,2}$	140.9478951190233
$A^{\beta}_{cu,1}$	4.934806872861141
$A^{\beta}_{cu,2}$	37.75976772070127
$A^{\beta}_{hu,1}$	13.265195513246809
$A_{1,1,1}$	71.04727671123747
$A_{1,2,2}$	69.0865541528717
$A_{2,1,2}$	140.9478951190233
$A_{cu,1}$	4.934806872861141
$A_{cu,2}$	37.75976772070127
$A_{hu,1}$	13.265195513246809
$dt_{1,1,1}$	32.696988254129245
$dt_{1,1,2}$	10.0
$dt_{1,2,2}$	81.90903523761222
$dt_{1,2,3}$	36.909035237612244
$dt_{2,1,2}$	18.090964762387785
$dt_{2,1,3}$	58.56822357179081
$dt_{cu,1}$	66.90903523761224
$dt_{cu,2}$	148.5682235717908
$dt_{hu,1}$	62.696988254129245
$f^{H}_{1,1,1}$	2.4318201700852766
$f^{H}_{1,2,2}$	10.0
$f^{H}_{2,1,2}$	11.18261971877969
$f^{C}_{1,1,1}$	15.0
$f^{C}_{1,2,2}$	13.0
$f^{C}_{2,1,2}$	10.452731979334096

Variable	Value
$RecLMTD_{1,1,1}$	0.05219625593706871
$RecLMTD_{1,2,2}$	0.017714501064838913
$RecLMTD_{2,1,2}$	0.029023162578148194
$RecLMTD_{cu,1}$	0.014610721030303163
$RecLMTD_{cu,2}$	0.009577839017992827
$RecLMTD_{hu,1}$	0.022543850648727472
$q_{1,1,1}$	680.9096476238775
$q_{1,2,2}$	1950.0
$q_{2,1,2}$	2428.635528564184
$q_{cu,1}$	169.09035237612238
$q_{cu,2}$	1971.364471435816
$q_{hu,1}$	490.4548238119387
$t^{(H)}_{1,1}$	650.0
$t^{(H)}_{1,2}$	581.9090352376122
$t^{(H)}_{1,3}$	386.9090352376122
$t^{(H)}_{2,1}$	590.0
$t^{(H)}_{2,2}$	590.0
$t^{(H)}_{2,3}$	468.5682235717908
$t^{(C)}_{1,1}$	617.3030117458708
$t^{(C)}_{1,2}$	571.9090352376122
$t^{(C)}_{1,3}$	410.0
$t^{(C)}_{2,1}$	500.0
$t^{(C)}_{2,2}$	500.0
$t^{(C)}_{2,3}$	350.0
$t^{H}_{1,1,1}$	370.0
$t^{H}_{1,2,2}$	386.9090352376123
$t^{H}_{2,1,2}$	372.82052062580465
$t^{C}_{1,1,1}$	617.3030117458708
$t^{C}_{1,2,2}$	500.0
$t^{C}_{2,1,2}$	642.3445711002439
$b^{H,\text{in}}_{1,1,1}$	1580.6831105554297
$b^{H,\text{in}}_{1,2,2}$	5819.090352376123
$b^{H,\text{in}}_{2,1,2}$	6597.745634080017

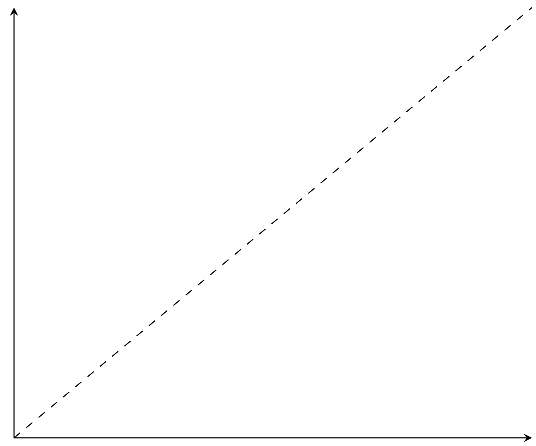
Variable	Value
$b^{H,\text{out}}_{1,1,1}$	899.7734629315523
$b^{H,\text{out}}_{1,2,2}$	3869.0903523761226
$b^{H,\text{out}}_{2,1,2}$	4169.110105515833
$b^{C,\text{in}}_{1,1,1}$	8578.635528564184
$b^{C,\text{in}}_{1,2,2}$	4550.0
$b^{C,\text{in}}_{2,1,2}$	4285.62011152698
$b^{C,\text{out}}_{1,1,1}$	9259.545176188061
$b^{C,\text{out}}_{1,2,2}$	6500.0
$b^{C,\text{out}}_{2,1,2}$	6714.2556400911635

3 *RecLMTD* Tangent Points

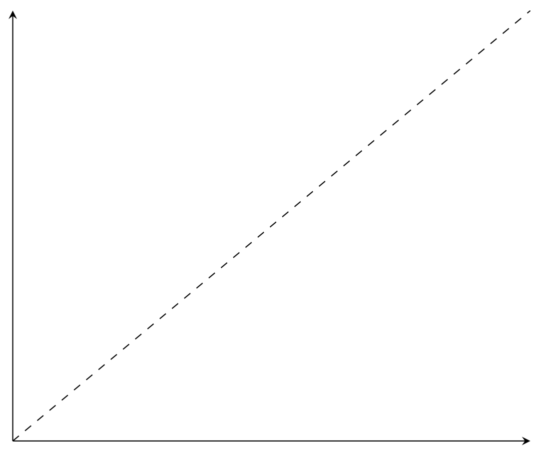
- $(1, 1, 1)$
  - Value: 0.052196
  - Error: 0.000000003103236
  - Relative error: 0.000000059453220
- $(1, 2, 2)$
  - Value: 0.017715
  - Error: 0.000000008416807
  - Relative error: 0.000000475136342
- $(2, 1, 2)$
  - Value: 0.029023
  - Error: 0.000000039460614
  - Relative error: 0.000001359623023
- $cu, 1$
  - Value: 0.014611
  - Error: 0.000000001393743
  - Relative error: 0.000000095391798
- $cu, 2$
  - Value: 0.009578
  - Error: 0.000000448417735
  - Relative error: 0.000046816065789
- $hu, 1$
  - Value: 0.022544
  - Error: 0.000000000328297
  - Relative error: 0.000000014562618



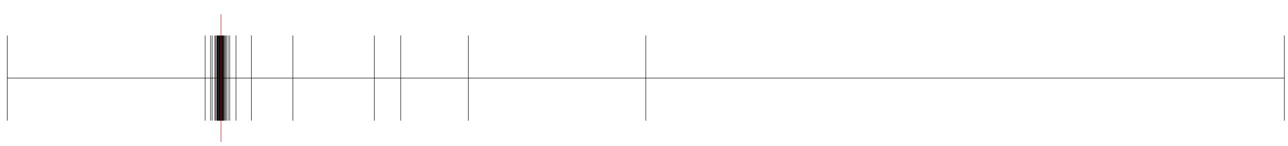
(a) *RecLMTD* - $(1, 1, 1)$



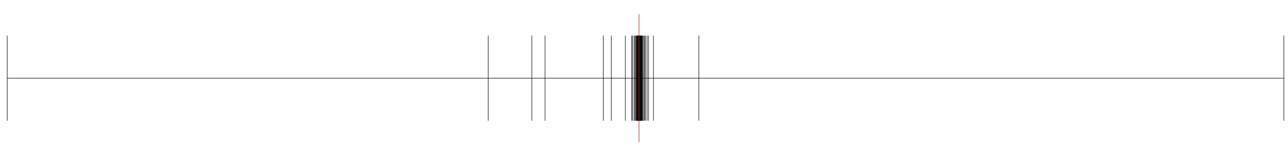
(b) *RecLMTD* - $(1, 2, 2)$



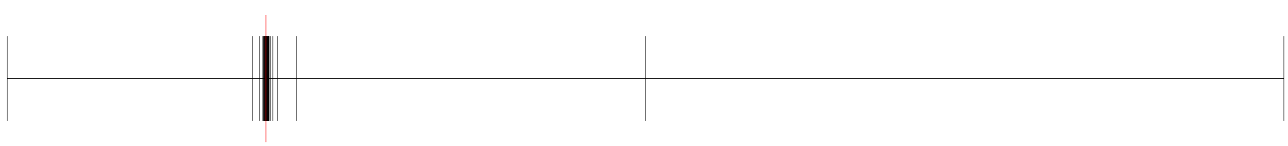
(c) *RecLMTD* - $(2, 1, 2)$



(a) *RecLMTD* - $cu, 1$



(b) *RecLMTD* - $cu, 2$



(a) *RecLMTD* - $hu, 1$

4 Balancing Breakpoints

4.1 $b^{\boldsymbol{H},\mathbf{in}}$

- $(1, 1, 1)$
  - Value: 1580.6831105554297
  - Absolute error: 0.0
  - Relative error: 0.0
- $(1, 2, 2)$
  - Value: 5819.090352376123
  - Absolute error: 0.0
  - Relative error: 0.0
- $(2, 1, 2)$
  - Value: 6597.745634080017
  - Absolute error: 0.0
  - Relative error: 0.0

4.2 $b^{\boldsymbol{H},\mathbf{out}}$

- $(1, 1, 1)$
  - Value: 899.7734629315523
  - Absolute error: 0.0
  - Relative error: 0.0
- $(1, 2, 2)$
  - Value: 3869.0903523761226
  - Absolute error: 0.0
  - Relative error: 0.0
- $(2, 1, 2)$
  - Value: 4169.110105515833
  - Absolute error: 0.0
  - Relative error: 0.0

4.3 $b^{\boldsymbol{C},\mathbf{in}}$

- $(1, 1, 1)$
  - Value: 8578.635528564184
  - Absolute error: 0.0
  - Relative error: 0.0
- $(1, 2, 2)$
  - Value: 4550.0
  - Absolute error: 0.0

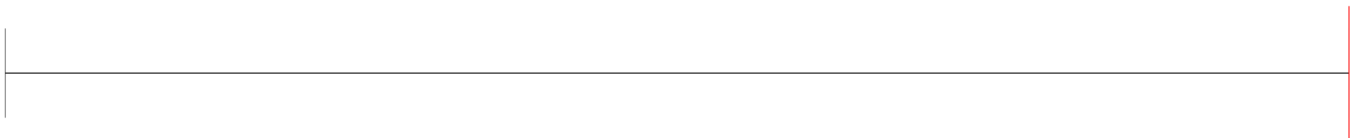
  - Relative error: 0.0
- $(2, 1, 2)$
  - Value: 4285.62011152698
  - Absolute error: 0.0
  - Relative error: 0.0

4.4 $b^{\mathbf{C,out}}$

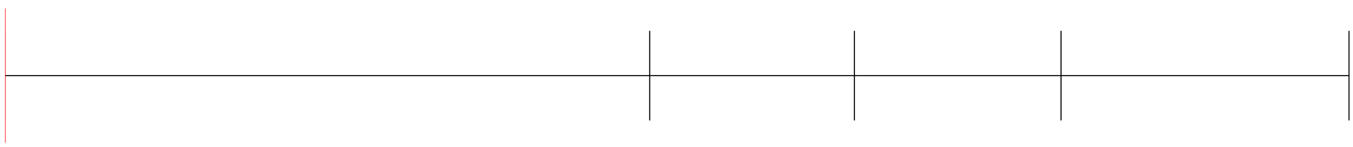
- $(1, 1, 1)$
  - Value: 9259.545176188061
  - Absolute error: 0.0
  - Relative error: 0.0
- $(1, 2, 2)$
  - Value: 6500.0
  - Absolute error: 0.0
  - Relative error: 0.0
- $(2, 1, 2)$
  - Value: 6714.2556400911635
  - Absolute error: 0.0
  - Relative error: 0.0

(a) $t^{(H)}$ - $(1, 1)$

(a) $t^{(H)}$ - $(1, 2)$

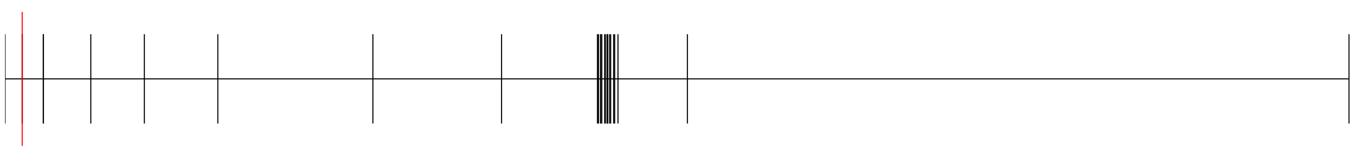


(a) $t^{(H)}$ - $(2, 2)$



(a) $t^{(H)}$ - $(1, 1, 1)$

(a) $t^{(H)}$ - $(1, 2, 2)$



(a) $t^{(H)}$ - $(2, 1, 2)$

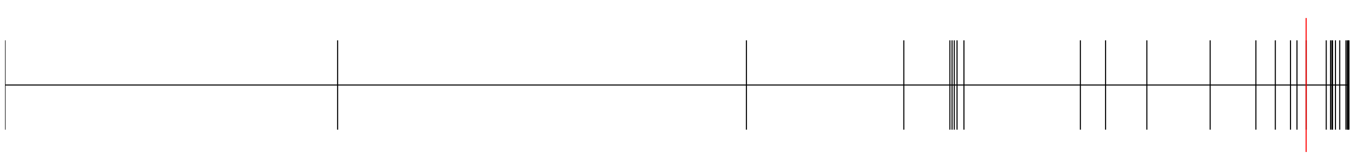
(a) $t^{(C)}$ - $(1, 2)$

(a) $t^{(C)}$ - $(2, 3)$

(a) $t^{(C)}$ - $(1,3)$

(a) $t^{(C)}$ - $(1,1,1)$

(a) $t^{(C)}$ - $(1,2,2)$



(a) $t^{(C)}$ - $(2,1,2)$

5 Area Breakpoints

5.1 A

- $(1, 1, 1)$
  - Value: 71.047277
  - Error: 0.034591763552882
  - Relative error: 0.000486646796083
- $(1, 2, 2)$
  - Value: 69.086554
  - Error: 0.000000000000057
  - Relative error: 0.000000000000001
- $(2, 1, 2)$
  - Value: 140.947895
  - Error: 0.025472458147078
  - Relative error: 0.000180689860680

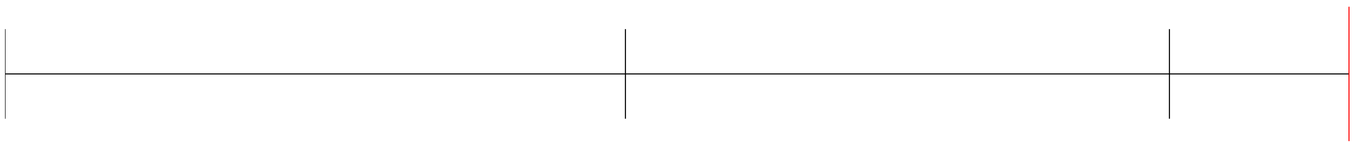
5.2 A_{CU}

- $cu, 1$
  - Value: 4.934807
  - Error: 0.006257062105226
  - Relative error: 0.001266339028918
- $cu, 2$
  - Value: 37.759768
  - Error: 0.003055385704265
  - Relative error: 0.000080909885780

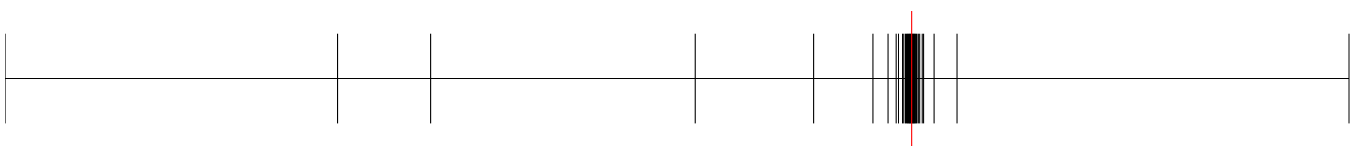
5.3 A_{HU}

- $hu, 1$
  - Value: 13.265196
  - Error: 0.002892844310340
  - Relative error: 0.000218030226539

(a) A^β - $(1,1,1)$

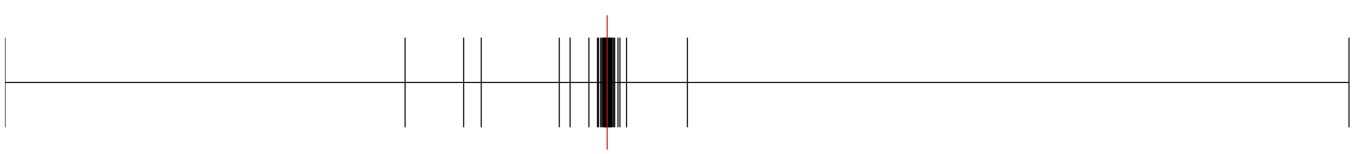


(a) A^β - $(1,2,2)$



(a) A^β - $(2,1,2)$

(a) A^β - $cu, 1$



(a) A^β - $cu, 2$

(a) A^β - $hu, 1$

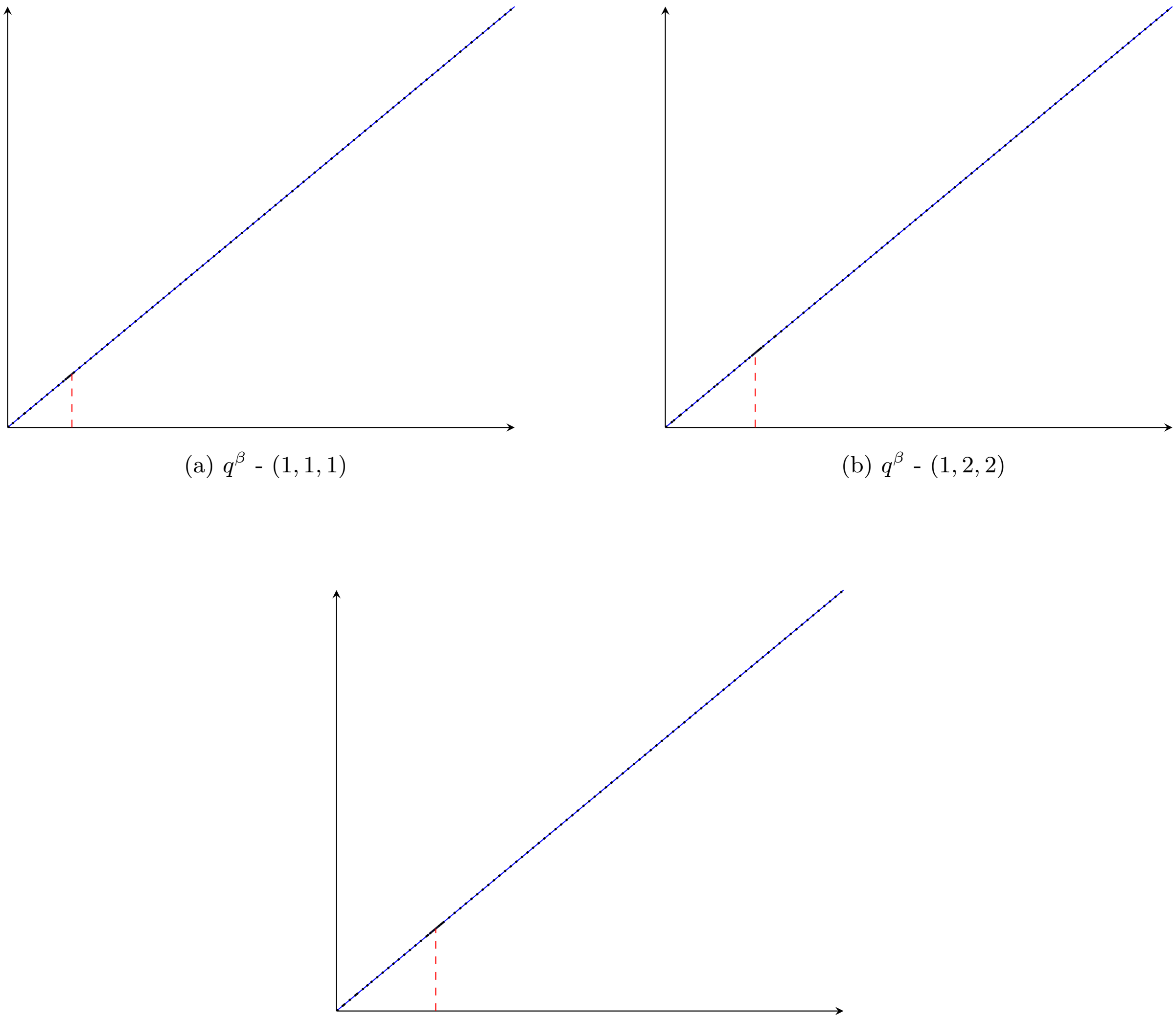
6 A^β Breakpoints

- $(1, 1, 1)$
  - q: 71.047277
  - $\hat{q}^\beta$: 71.047277
  - q^β: 71.047277
  - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.42777867315647, 69.49758876855047, 69.89902342903967, 70.11420238741596, 70.22290217272871, 70.30181758862284, 70.46247894104941, 70.50425945122109, 70.58322282660124, 70.60541450524272, 70.65581083979713, 70.67982582933766, 70.76867866839005, 70.83181816751056, 70.8571173980136, 70.94547904612048, 70.95870595008942, 71.00935653192329, 71.03465240716494, 71.1226084153824, 71.18527282143235, 71.21162487917152, 71.22933662920877, 71.2616536557244, 71.30011482813455, 71.33750386361461, 71.3743618177527, 71.52154532022675, 71.62767096549581, 71.6637831223933, 71.88123217857378, 72.01156079815767, 72.36161285231599, 560.0]
  - Error: 0.000000000000000
  - Relative error: 0.000000000000000
- $(1, 2, 2)$
  - q: 69.086554
  - $\hat{q}^\beta$: 69.086554
  - q^β: 69.086554
  - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, 66.74637189255778, 67.27673099173637, 67.60071888621576, 67.91344539650363, 68.10969007738701, 68.20353230192663, 68.3940471550051, 68.48329505760876, 68.57240847861726, 68.66676327863097, 68.76047282726, 68.85281320938876, 68.94418284540035, 69.0314457108091, 69.13316279873226, 69.22655548799992, 69.3178258940278, 69.34502840170047, 69.41355730879985, 69.51025044804472, 69.60484346150977, 69.70052578983292, 69.93657924014246, 70.08419437796795, 70.18046341267424, 70.37078617087542, 70.56626459266522, 70.94081287188642, 71.26369181826138, 72.25348351595485, 72.87506706799363, 74.6496383881616, 84.1468297226002, 390.0]
  - Error: 0.000000000000000
  - Relative error: 0.000000000000000
- $(2, 1, 2)$
  - q: 140.947895
  - $\hat{q}^\beta$: 140.947895
  - q^β: 140.947895
  - Breakpoints: [0, 9.898970403313276, 28.333333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 127.07690270456857, 128.52021081477858, 131.69760457629607, 132.12228195956155, 132.5928424015674, 135.25996987589937, 135.81963163113545, 136.73553825185095, 137.02923097052596, 137.63821724432228, 138.55705332155773, 138.86359281280824, 139.27702937706263, 139.47623233772285, 140.0019654200325, 140.31685261310218, 140.7008934199032, 140.73764915547463, 141.47857504067568, 141.79783607898793, 142.2268253113294, 142.4338700260237, 142.98255202451594, 143.30274961189664, 143.73343297342473, 144.42095395765313, 144.67005099128556, 144.70843093515643, 147.02575214945074, 147.07009017600456, 147.48374761404892, 149.84689443310864, 150.84969099235698, 720.0]
  - Error: 0.000000000000000
  - Relative error: 0.000000000000000
- $cu, 1$

- q: 4.934807
- $\hat{q}^\beta$: 4.934807
- q^β: 4.934807
- Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 3.9948823402911255, 4.0691182452509045, 4.269280092716329, 4.432280503220532, 4.440201431981878, 4.600843803756838, 4.613047202566896, 4.701079869478363, 4.707108999269609, 4.75409580038526, 4.8073435862836025, 4.813467676447643, 4.825601156120121, 4.844843256177314, 4.849719301830286, 4.882089573558703, 4.888121952381175, 4.9192700692457025, 4.956260392645724, 4.962276317638335, 4.974201802143407, 4.983938626310074, 4.993068227518925, 4.993351789275153, 5.030173826900646, 5.0359974771272995, 5.0418124100090695, 5.066729567155887, 5.081618324442152, 5.0892956597199035, 5.195781532076625, 5.3386847474643435, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]
- Error: 0.000000000000000
- Relative error: 0.000000000000000

- $cu, 2$
  - q: 37.759768
  - $\hat{q}^\beta$: 37.759768
  - q^β: 37.759768
  - Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, 37.21616748032837, 37.40128299925594, 37.443498367229196, 37.47875108132638, 37.52720514437563, 37.554741031910254, 37.612770977560544, 37.63325504358228, 37.65308365461216, 37.65435035996605, 37.6551299822246, 37.682669717644174, 37.68654427850372, 37.711317605988725, 37.71812813120871, 37.738762662315665, 37.7631578546152, 37.76995088018704, 37.78573239590169, 37.791030068986316, 37.81486687126635, 37.821651654621526, 37.84304537613103, 37.87287854392874, 37.88148462263016, 37.9018730666985, 37.90741115359579, 37.9454585315849, 37.95085342166671, 38.035748486883605, 38.10872593446926, 38.21652191748154, 38.224970554765775, 38.88483572219945, 285.4001551947793]
  - Error: 0.000000000000000
  - Relative error: 0.000000000000000

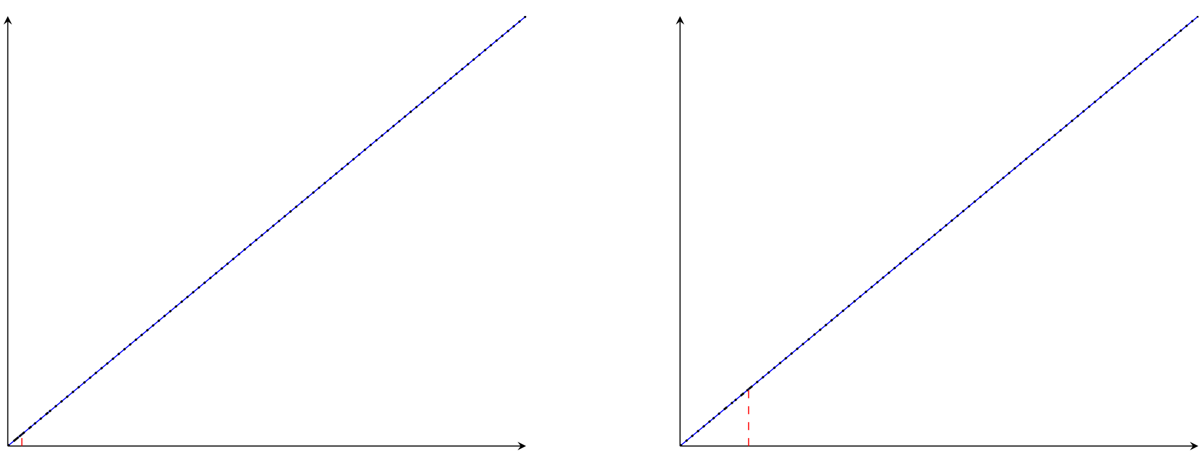
- $hu, 1$
  - q: 13.265196
  - $\hat{q}^\beta$: 13.265196
  - q^β: 13.265196
  - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 12.917705220652834, 13.064750414488246, 13.104034300095861, 13.134779411775817, 13.161020122653593, 13.189852638760229, 13.192252946941991, 13.195607617969486, 13.21041298113453, 13.211215176722128, 13.213749911872785, 13.218222920629291, 13.224123753786348, 13.22717920580809, 13.23798359102618, 13.238585544475612, 13.244985069124171, 13.251789114535772, 13.265751604070672, 13.266498143734903, 13.269097009439808, 13.279580305153486, 13.293477718720517, 13.294574825457861, 13.307344891293862, 13.30871858348944, 13.321736298873567, 13.32330648750218, 13.32663103908035, 13.341988022860814, 13.346967780023082, 13.363827419897227, 13.411355634100326, 13.439500729482463, 13.468285806082731, 13.498843422507207, 14.09178614751222, 237.3002543523117]
  - Error: 0.000000000000000
  - Relative error: 0.000000000000000



(a) q^β - $(1,1,1)$

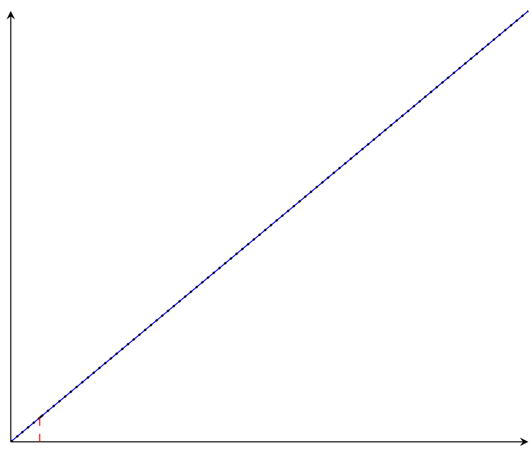
(b) q^β - $(1,2,2)$

(c) q^β - $(2,1,2)$



(a) q^β - $cu, 1$

(b) q^β - $cu, 2$



(a) q^β - $hu, 1$

7 Other variables

Variable	Value
$z_{1,1,2}$	0.0
$z_{1,2,1}$	-0.0
$z_{2,1,1}$	-0.0
$z_{2,2,1}$	-0.0
$z_{2,2,2}$	0.0
$z_{hu,2}$	-0.0
$A^{\beta}_{1,1,2}$	0.0
$A^{\beta}_{1,2,1}$	0.0
$A^{\beta}_{2,1,1}$	0.0
$A^{\beta}_{2,2,1}$	0.0
$A^{\beta}_{2,2,2}$	0.0
$A^{\beta}_{hu,2}$	0.0
$A_{1,1,2}$	0.0
$A_{1,2,1}$	0.0
$A_{2,1,1}$	0.0
$A_{2,2,1}$	0.0
$A_{2,2,2}$	0.0
$A_{hu,2}$	0.0
$dt_{1,1,3}$	10.0
$dt_{1,2,1}$	280.0
$dt_{2,1,1}$	24.166666666666668
$dt_{2,2,1}$	90.0
$dt_{2,2,2}$	220.0
$dt_{2,2,3}$	240.0
$dt_{hu,2}$	180.0
$f^{H}_{1,1,2}$	0.0
$f^{H}_{1,2,1}$	7.568179829914723
$f^{H}_{2,1,1}$	20.0
$f^{H}_{2,2,1}$	0.0
$f^{H}_{2,2,2}$	8.81738028122031
$f^{C}_{1,1,2}$	4.5472680206659035
$f^{C}_{1,2,1}$	13.0
$f^{C}_{2,1,1}$	0.0

Variable	Value
$f^{C}_{2,2,1}$	0.0
$f^{C}_{2,2,2}$	0.0
$RecLMTD_{1,1,2}$	0.1
$RecLMTD_{1,2,1}$	0.01607047381606835
$RecLMTD_{2,1,1}$	0.1
$RecLMTD_{2,2,1}$	0.00608
$RecLMTD_{2,2,2}$	0.004340277777777778
$RecLMTD_{hu,2}$	0.005549329826813255
$q_{1,1,2}$	0.0
$q_{1,2,1}$	0.0
$q_{2,1,1}$	0.0
$q_{2,2,1}$	0.0
$q_{2,2,2}$	0.0
$q_{hu,2}$	0.0
$t^{H}_{1,1,2}$	418.8196134269507
$t^{H}_{1,2,1}$	650.0
$t^{H}_{2,1,1}$	590.0
$t^{H}_{2,2,1}$	590.0
$t^{H}_{2,2,2}$	590.0
$t^{C}_{1,1,2}$	452.0803398134609
$t^{C}_{1,2,1}$	500.0
$t^{C}_{2,1,1}$	410.0
$t^{C}_{2,2,1}$	350.0
$t^{C}_{2,2,2}$	370.0454403499536
$b^{H,\text{in}}_{1,1,2}$	0.0
$b^{H,\text{in}}_{1,2,1}$	4919.31688944457
$b^{H,\text{in}}_{2,1,1}$	11800.0
$b^{H,\text{in}}_{2,2,1}$	0.0
$b^{H,\text{in}}_{2,2,2}$	5202.254365919983
$b^{H,\text{out}}_{1,1,2}$	0.0
$b^{H,\text{out}}_{1,2,1}$	4919.31688944457
$b^{H,\text{out}}_{2,1,1}$	11800.0
$b^{H,\text{out}}_{2,2,1}$	0.0

Variable	Value
$b^{H,\text{out}}_{2,2,2}$	5202.254365919983
$b^{C,\text{in}}_{1,1,2}$	1864.3798884730204
$b^{C,\text{in}}_{1,2,1}$	6500.0
$b^{C,\text{in}}_{2,1,1}$	0.0
$b^{C,\text{in}}_{2,2,1}$	0.0
$b^{C,\text{in}}_{2,2,2}$	0.0
$b^{C,\text{out}}_{1,1,2}$	1864.3798884730204
$b^{C,\text{out}}_{1,2,1}$	6500.0
$b^{C,\text{out}}_{2,1,1}$	0.0
$b^{C,\text{out}}_{2,2,1}$	0.0
$b^{C,\text{out}}_{2,2,2}$	0.0